Higher-Order Modular Attention: Domain-Specific Transformer Architectures for Protein Sequence Modeling

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Abstract

Since the Transformer, many attention variants have improved efficiency and context modeling, yet most still rely on pairwise token interactions, which can limit the representation of higher-order dependencies that arise in biological sequences. We propose a higher-order attention mechanism that captures multi-token interactions beyond standard dot-product attention while remaining scalable through structured sparsity. Building on this mechanism, we introduce six Transformer architectures that span local-to-global dependency regimes and support both classification and regression settings. To enable controlled comparisons, we implement a unified, modular Transformer pipeline where attention is a plug-and-play component, allowing existing and custom attention modules to be evaluated under an identical training and evaluation protocol. Across TAPE protein benchmarks, our models achieve consistent improvements over strong Transformer baselines, with the largest gains on tasks requiring richer contextual interactions, while maintaining competitive efficiency.

1 Introduction

- make sure to mention that the secondary structure a lot of times depends on the local triple interactions.

Biological sequences encode functional behaviour through dependencies that extend beyond simple pairwise relations. Across proteins, RNAs, and genomic regulatory regions, coordinated patterns of variation and structure frequently arise from the joint contribution of three or more positions. These higher order interactions influence stability, catalytic function, evolutionary adaptation, and regulatory control, and have been documented in a wide range of molecular systems.

In proteins, three way epistasis plays a central role in shaping structural and functional outcomes. Classical catalytic triads demonstrate that enzymatic activity depends on a coordinated triplet rather than independent residue pairs [**?**]. Deep mutational scanning has shown that many fitness effects arise from combinations of three interacting residues, indicating that higher order couplings are pervasive in protein evolution [**? ?**]. Coevolutionary analyses further reveal that residue triplets can mediate long range allosteric communication and cooperative stabilisation of tertiary structures [**? ?**]. These findings cannot be explained by pairwise statistics alone.

Triplet interactions are also common in viral and microbial genomes. In rapidly evolving pathogens such as influenza and SARS–CoV–2, the functional effect of a mutation often depends on the presence

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of two additional mutations, forming synergistic sets that modulate receptor binding, antigenicity, or immune escape [**? ?**]. Coordinated mutation triplets in HIV shape tropism and drug susceptibility [**?**]. Similar patterns appear in bacteria, where three way epistasis guides antibiotic resistance trajectories and compensatory adaptation [**?**].

Regulatory DNA and RNA sequences also rely on cooperative multi positional logic. Enhancer activity frequently depends on specific combinations of transcription factor binding sites whose effect emerges only when all required motifs appear in the correct arrangement [**? ?**]. RNA structures contain coordinated nucleotide triplets that stabilise local folds, pseudoknots, or catalytic centres, and the phenotypic effect of a substitution often depends on the joint state of several additional nucleotides [**?**]. These examples reinforce that combinatorial dependencies are a defining feature of non coding regions.

The prevalence of higher order dependencies has direct statistical implications. From a functional ANOVA perspective, any sequence–to–function mapping $f(x_1, \ldots, x_N)$ admits a decomposition

$$f = \sum_i f_i(x_i) + \sum_{i<j} f_{ij}(x_i, x_j) + \sum_{i<j<k} f_{ijk}(x_i, x_j, x_k) + \cdots ,$$

where f_{ijk} represents the pure three way interaction that remains after all main and pairwise contributions are removed. Conventional attention mechanisms approximate only the first two components through their bilinear structure, leaving f_{ijk} to be reconstructed indirectly through depth and nonlinearities, which is inefficient when higher order terms are substantial.

Information theory arrives at the same conclusion. True triplet dependence corresponds to nonzero interaction information,

$$I(X_i; X_j; X_k) = I(X_i; X_j) - I(X_i; X_j \mid X_k),$$

which quantifies whether the relationship between two positions changes once a third is taken into account. Biological sequences routinely display significant interaction information across triplets, meaning that models restricted to pairwise terms necessarily fail to extract all the information present in the data. Introducing an explicit third order interaction term therefore provides a principled way to capture structure that pairwise attention cannot represent.

{Maybe to be included: Finally, the third order component introduced in our attention mechanism can be understood as a structured analogue of a third order Volterra kernel or cubic polynomial transform in the latent representation. Classical systems theory shows that including explicit second and third order kernels increases the ability of a model to approximate nonlinear functionals with far fewer parameters than attempting to emulate the same interactions through compositions of linear operations. In this sense, the three way attention term provides a principled inductive bias that directs the model to search for cubic interaction patterns between embedding dimensions at three distinct sequence positions. This paragraph can be later linked to LoRA-like method of adding the 3D method on top of 2d model to improve performance instead of increasing the number of parameters in the dot-product attention. this can have the promise of instead of increasing the number of parameters, we are changing the way the model is extracting information. }

Taken together, these biological and statistical considerations highlight a recurring principle: many meaningful sequence features arise from higher order dependencies that cannot be represented adequately by pairwise models. Standard self attention mechanisms rely on a bilinear comparison between two positions and therefore lack an explicit mechanism for capturing coordinated triplets. This limits their ability to detect multi residue stabilisation motifs, epistatic mutation combinations, and regulatory triplets that jointly determine functional behaviour.

To address this gap, we introduce an attention mechanism that incorporates both a conventional pairwise term and an explicit third order interaction term capable of assigning importance to triplets of positions. This formulation directly targets the higher order components of the functional decomposition and provides the capacity to represent nonzero interaction information. Applied to the TAPE secondary structure prediction benchmark, the model consistently outperforms the standard two dimensional attention across multiple test conditions, demonstrating that explicit modelling of three way interactions yields measurable benefits in tasks where biological function is shaped by coordinated dependencies among multiple positions. To help answer this question, we introduce the Tasks Assessing Protein Embeddings (TAPE), which to our knowledge is the first attempt at systematically evaluating semi-supervised learning on protein sequences. TAPE includes a set of five

biologically relevant supervised tasks that evaluate the performance of learned protein embeddings across diverse aspects of protein understanding. We choose our tasks to highlight three major areas of protein biology where self-supervision can facilitate scientific advances: structure prediction, detection of remote homologs, and protein en- gineering. Improvement on these tasks range in application, including designing new antibodies [8] (for ss)[Timothy A Whitehead, Aaron Chevalier, Yifan Song, Cyrille Dreyfus, Sarel J Fleishman, Cecilia De Mattos, Chris A Myers, Hetunandan Kamisetty, Patrick Blair, Ian A Wilson, et al. Optimization of affinity, specificity and function of designed influenza inhibitors using deep sequencing. Nature biotechnology, 30(6):543, 2012.]. We show that self-supervised pretraining improves performance for almost all models on all downstream tasks. Interestingly, performance for each architecture varies significantly across tasks, highlighting the need for a multi-task benchmark such as ours. We also show that non-deep alignmentbased features [13–16] outperform features learned via self-supervision on secondary structure and contact prediction, while learned features perform significantly better on remote homology detection. Our results demonstrate that self-supervision is a promising paradigm for protein modeling but considerable improvements need to be made before self-supervised models can achieve breakthrough performance. All code and data for TAPE are publically available1, and we encourage members of the machine learning community to participate in these exciting problems.

1.1 Background

Proteins are linear chains of amino acids connected by covalent bonds.Beyond its encoding as a sequence (x1; : : : ; xL), a protein has a 3D molecular structure. The different levels of protein structure include primary (amino acid sequence), secondary (local features), and tertiary (global features). Understanding how primary sequence folds into tertiary structure is a fundamental goal of biochemistry. [C B Anfinsen, E Haber, M Sela, F H White, and Jr. The kinetics of formation of native ribonuclease during oxidation of the reduced polypeptide chain. Proceedings of the National Academy of Sciences of the United States of America, 47(9):1309–14, 1961.] Protein secondary structure can be calculated based on its atoms' 3D coordinates once the protein's 3D structure is solved using X-ray crystallography or NMR. However, X-ray or NMR is expensive. Ideally, we would like to predict the secondary structure of a protein based on its primary sequence directly. SS is an important feature for understanding the function of a protein, especially if the protein of interest is not evolutionarily related to proteins with known structure [36][Michael Schantz Klausen, Martin Closter Jespersen, Henrik Nielsen, Kamilla Kjaergaard Jensen, Vanessa Isabell Jurtz, Casper Kaae Soenderby, Morten Otto Alexander Sommer, Ole Winther, Morten Nielsen, Bent Petersen, et al. Netsurfp-2.0: Improved prediction of protein structural features by integrated deep learning. Proteins: Structure, Function, and Bioinformatics, 2019.] . tools are very commonly used to create richer input features for higher-level models [37][Alexey Drozdetskiy, Christian Cole, James Procter, and Geoffrey J. Barton. JPred4: a protein secondary structure prediction server. Nucleic Acids Research, 43(W1):W389–W394, 2015.]. [might help: James A Cuff and Geoffrey J Barton. Evaluation and improvement of multiple sequence methods for protein secondary structure prediction. Proteins: Structure, Function, and Bioinformatics, 34(4):508–519, 1999.]

1.2 Related Work

cite the vanilla transformer model then the block attention idea give three references then the linear attention method. also cite the meta 3D model and the other reference that is similar to our work. in terms of related work, we must also give explanations to the tape leader board and the winning models at the top so that later we can say our method surpassed top performance in tape leader board. or we should add these to the related work and skip intro here. E and F projections are specific to each layer. the projection k summerizes legnth of sequence into k dimention. as standard multi-head self-attention, overlapping block-wise 2D attention with block length ℓ and stride s, and linear 2D attention via a projection-style low-rank approximation that compresses the sequence dimension to k,

2 Experimental Setup and Datasets

We use a unified Transformer pipeline in which the attention module is the only varying component, enabling controlled comparisons between HOMA and attention baselines. We consider global multi-head self-attention (Pairwise-2D), overlapping block-wise attention (Blockwise-2D), and Linformer-

style low-rank attention (Linear-2D), and evaluate HOMA with triadic window sizes $w \in \{3, 5, 7\}$. Model scale is matched as closely as possible across variants; any remaining differences are attention-specific (Table 2). Experiments run on a single NVIDIA A100-SXM4-80GB GPU. To separate model compute from data-pipeline overhead, we report compute-only throughput (token-positions/second) using CUDA synchronization around each optimization step, and provide end-to-end throughput in the supplementary material. We also report peak GPU memory allocation and use fixed seeds with early stopping based on validation performance.

Datasets, Tasks, and Metrics

We evaluate on three supervised datasets from the TAPE benchmark, spanning residue-level classification, sequence-level regression under combinatorial mutation, and local sensitivity to single mutations [1]. We describe each dataset below.

Secondary Structure. Secondary structure prediction is a residue-level token classification task. Given $x = (x_1, x_2, \ldots, x_L)$, the model predicts $y_i \in \{\text{helix}, \text{strand}, \text{coil}\}$ for each residue x_i. Training and validation data are sourced from Klausen et al. [2]. We evaluate on CB513 [3], CASP12 [4], and TS115 [5], reporting Q3 accuracy and F1 over non-padded residues. Models are trained with token-level cross-entropy, using an ignore index of -100 at padded positions. The dataset contains 8,678 training sequences and 2,170 validation sequences.

Fluorescence. Fluorescence is a sequence-level regression task over mutated green fluorescent protein variants. The model predicts $y \in \mathbb{R}$ corresponding to log-fluorescence intensity measured by Sarkisyan et al. [6]. The training split contains variants within Hamming distance 3 of the parent sequence, while the test split contains variants with at least 4 mutations [1]. The dataset contains 21,446 training sequences, 5,362 validation sequences, and 27,217 test sequences.

Stability. Stability is a sequence-level regression task over designed protein variants. The model predicts $y \in \mathbb{R}$ as a proxy for thermodynamic stability derived from high-throughput folding measurements [7]. The training split aggregates sequences from multiple rounds of experimental design, while the test split consists of Hamming distance of 1 from top designs. The dataset contains 53,614 training sequences, 2,512 validation sequences, and 12,851 test sequences.

For the regression tasks, we train with mean squared error and report Spearman rank correlation ρ on the test set [1].

Preprocessing and sequence handling. All datasets use residue-level tokenization with the TAPE IUPAC tokenizer. The tokenizer defines a 30-token vocabulary consisting of five control symbols and a 25-character residue alphabet, where the residue alphabet comprises 20 standard amino acids, two non-standard amino acids selenocysteine and pyrrolysine, two ambiguity codes, and one unknown-residue symbol [8, 9]. To support a unified pipeline, sequences are padded or truncated to a maximum length of 512 using the `<pad>` token, and padded positions are excluded from attention and evaluation via masking.

3 Results

Table 2 summarizes performance on CASP12 using F1 and accuracy, and reports Spearman correlation ρ on Fluorescence and Stability. Results on the remaining secondary-structure test sets TS115 and CB513 are provided in the supplementary material.

Among the baseline attention modules, Blockwise-2D is consistently strongest across all three benchmarks. On CASP12, it raises accuracy from 0.5582 to 0.6368 (a 14.08% relative gain) and F1 from 0.5257 to 0.6228 (an 18.47% gain) compared to Pairwise-2D. For Fluorescence, Blockwise-2D achieves $\rho = 0.6998$ versus 0.6647 with Pairwise-2D, corresponding to a 5.28% relative improvement. On Stability, the correlation increases from $\rho = 0.6221$ to $\rho = 0.6509$, a 4.63% gain. Overall, these comparisons indicate that restricting attention to overlapping local blocks yields robust improvements for both residue-level secondary-structure prediction and sequence-level property prediction.

Model	Attention	Key hyperparameters	Params (SS)	Params (F&S)
(A) Baselines				
Pairwise-2D	Global MHSA	–	≈25.5M	≈20.9M
Blockwise-2D	Overlapping block MHSA	ℓ=b, s=15	≈25.5M	≈21.3M
Linear-2D	Linformer-style attention	k=50	≈26.1M	≈22.1M
(B) Proposed				
HOMA	Pairwise + windowed Triadic	ℓ=b, s=15, r=8	≈25.9M	≈21.5M
(C) Shared settings				
d_{model}	SS / F&S		512	256
Layers	SS / F&S		12	12
Heads	SS / F&S		8	8
FFN dim	SS / F&S		1024	128
Dropout	SS / F&S		0.4	0.4
LR	SS / F&S		10^{-4}	5×10^{-5}

Table 1: Model variants and configurations. Will be rewritten. SS denotes Secondary Structure; F&S denotes Fluorescence and Stability. l stands for length of blocks, s is the stride. w is window size in each block for triadic attention. r stands for the rank of the U matrix. rank is per layer. all models share the same backbone; only attention module changes; parameter counts shown. note, remove the ss and F&S here.

In contrast, Linear-2D is less effective on tasks that depend on local structural detail. It attains CASP12 accuracy 0.5458 and Stability correlation $\rho = 0.5439$, both below Pairwise-2D and Blockwise-2D. Its performance on Fluorescence is more competitive: Linear-2D reaches $\rho = 0.6821$, slightly surpassing Pairwise-2D ($\rho = 0.6647$, a 2.62% relative gain) but still falling short of Blockwise-2D.

HOMA augments the Blockwise-2D backbone with an additional triadic interaction pathway. On CASP12 and Stability, the best configuration uses window size $w = 5$, reaching CASP12 accuracy 0.6588 and F1 0.6338, which correspond to relative gains of 3.45% in accuracy and 1.77% in F1 over Blockwise-2D. On Stability, the same setting attains $\rho = 0.7152$, improving over Blockwise-2D by 9.88%. For Fluorescence, the strongest HOMA variant uses $w = 7$ and obtains $\rho = 0.7388$, a 5.57% relative gain compared to Blockwise-2D. These results show that enriching local pairwise attention with triadic interactions systematically improves performance, with particularly pronounced gains on Stability.

Performance of HOMA attention models seem to depend on the window sizes. Relative to Blockwise-2D, $w = 3$ yields modest gains of 0.85% on CASP12 accuracy, 0.26% on Fluorescence, and 2.11% on Stability, while CASP12 F1 remains essentially unchanged. Increasing the window to $w = 5$ improves over $w = 3$ across all benchmarks, raising CASP12 accuracy by 2.58% and F1 by 1.80%, increasing Fluorescence by 1.43%, and increasing Stability by 7.61%. Expanding further to $w = 7$ improves Fluorescence by 3.82% relative to $w = 5$, but reduces CASP12 accuracy by 1.12%, reduces CASP12 F1 by 0.77%, and reduces Stability by 0.64% relative to $w = 5$.

Efficiency and resource trade-offs.

Figure 1a shows the validation loss over the first six epochs on the secondary-structure task, comparing the convergence behaviour of the baseline models to the HOMA variants. The losses for Pairwise-2D and Linear-2D quickly plateau and remain the highest curves in the plot, whereas the localized attention mechanisms (BLOCKWISE-2D and HOMA with $w \in \{3, 5, 7\}$) continue to decrease throughout training and achieve lower validation loss. This pattern suggests that explicitly modeling local structure improves both optimization dynamics and the model's ability to extract useful information from the sequence.

Figures 1b and 1c report compute-only throughput and peak GPU memory allocation, respectively, for HOMA and the baseline attention mechanisms on CASP12 and Fluorescence. Baseline variants follow expected efficiency trends: blockwise and linear attention improve throughput and reduce peak memory relative to full pairwise attention, with linear attention achieving the highest throughput and lowest memory footprint across both tasks; these serve primarily as reference points to contextualize the cost of higher-order modeling.

Model	CASP12		Fluorescence	Stability
	F1	Acc	ρ	ρ
Baseline attention				
Pairwise-2D	0.5257	0.5582	0.6647	0.6221
Blockwise-2D	0.6228	0.6368	0.6998	0.6509
Linear-2D	0.5466	0.5458	0.6821	0.5439
Augmented attention (HOMA)				
HOMA ($w = 3$)	0.6226	0.6422	0.7016	0.6646
HOMA ($w = 5$)	**0.6338**	**0.6588**	0.7116	**0.7152**
HOMA ($w = 7$)	0.6289	0.6514	**0.7388**	0.7106

Table 2: **Performance on TAPE.** Will be rewritten. We report F1/accuracy on CASP12 (secondary structure) and Spearman correlation (ρ) on Fluorescence and Stability for Transformer variants with different attention modules.

In contrast, HOMA exhibits a clear efficiency–expressivity trade-off. Compute throughput decreases monotonically as the local attention window size increases: on CASP12, throughput falls from roughly 3.1×10^4 token-positions/s at w=3 to 1.5×10^4 at w=7, with a similar pattern on Fluorescence. This reduction reflects the additional cost of constructing and processing higher-order local interaction tensors as the neighborhood expands. Peak GPU memory shows a complementary trend, rising from 14.6 GB to 38.9 GB on CASP12 and from 8.3 GB to 20.8 GB on Fluorescence as w grows. This increase is driven by the explicit materialization of intermediate higher-order representations in the mixed attention pathway.

Overall, these results demonstrate that HOMA introduces a controllable computational overhead in exchange for increased representational capacity over structured local interactions. The attention window size provides an explicit knob that allows practitioners to balance compute and memory budgets against modeling expressivity under fixed hardware constraints.

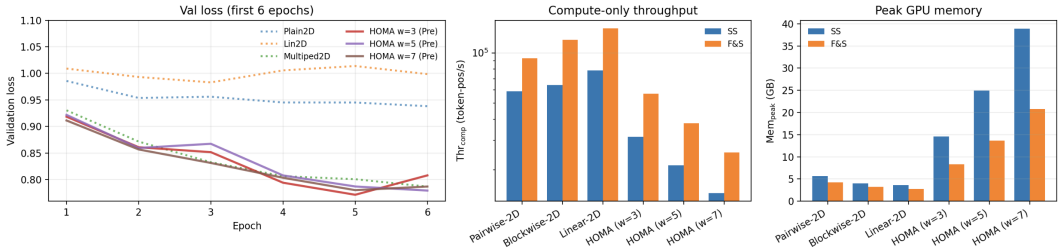

Figure 1: Will be rewritten once finalized. this equals the batch size times the fixed padded sequence length

4 Ablations and Analysis

We perform targeted ablation studies to isolate the contribution of three design and training choices in HOMA: (i) the low-rank factorization used to parameterize the U-matrix, (ii) initialization and optimization strategies for the pairwise (2D) attention backbone via pretraining and freezing, and (iii) the maximum sequence length used during training. All ablations are conducted on the TAPE secondary structure task, and we report accuracy@3 on the CASP12, TS115, and CB513 test sets. Figure 2 summarizes the resulting trends across datasets.

Rank ablation (Fig. 3A). We vary the rank used to factorize the U-matrix and observe a consistent advantage for the full-rank parameterization. Training from scratch with full U attains 66.6% on CASP12, 68.2% on TS115, and 65.5% on CB513, while the best low-rank setting in each case reaches 65.1% (rank-8) on CASP12, 67.9% (rank-8) on TS115, and 65.0% (rank-8) on CB513. On CASP12, accuracy decreases monotonically as rank is reduced from 128 to 8 (65.6%, 65.4%, 65.1%),

yielding a 1.5-point drop relative to full-rank. In contrast, TS115 and CB513 are less sensitive to aggressive factorization: rank-8 remains within 0.3 and 0.5 points of full-rank, respectively, and slightly exceeds rank-32 and rank-128 (TS115: 67.9% vs. 67.7%/67.7%; CB513: 65.0% vs. 64.8%/64.4%). This pattern suggests that while CASP12 benefits from higher-rank capacity, a compact rank-8 approximation can capture most of the useful structure on TS115 and CB513.

Pretraining and freezing of pairwise attention (Fig. 3B). We next ablate the training procedure by varying how attention parameters are initialized and optimized. In the first setting, both pairwise and triadic attention parameters are randomly initialized and trained end-to-end within HOMA (2D from scratch). In the second setting, we first train a model with pairwise (2D) attention, then transfer the learned pairwise attention parameters into HOMA as initialization; we either continue updating these transferred parameters during HOMA training (pretrained, no freeze) or keep them fixed to reduce compute and memory overhead (pretrained, frozen). Figure 3B shows that freezing the transferred pairwise weights consistently underperforms the other strategies, yielding 64.5% on CASP12, 65.5% on TS115, and 62.6% on CB513. Allowing the transferred weights to continue training largely closes the gap to training from scratch: on CASP12, HOMA with 2D from scratch achieves 65.9% compared to 65.1% for pretrained, no freeze (a 0.8–0.9 point advantage for scratch), while on TS115 the two are nearly identical (68.0% vs. 67.9%). On CB513, however, pretrained initialization with continued training for pairwise attention weights provides a clear benefit, reaching 68.0% versus 65.0% for training from scratch (a 3.0-point improvement). Overall, these results indicate that transferred pairwise attention can serve as an effective initialization for HOMA when it is allowed to adapt during training, whereas freezing the transferred parameters limits adaptation to the triadic pathway and leads to systematic accuracy drops despite potential efficiency gains.

Effect of maximum sequence length (Fig. 3C). Our third ablation study examines how the maximum sequence length used during training affects secondary-structure accuracy. We compare training with sequences truncated/padded to lengths 256, 512, and 1024. Across all three test sets, using length 256 yields the lowest performance, consistent with information loss from aggressive truncation (63.5% on CASP12, 66.3% on TS115, and 63.6% on CB513). Increasing the limit to length 512 provides the best overall results, reaching 65.1% on CASP12, 67.9% on TS115, and 68.0% on CB513. Extending to length 1024 improves over length 256 but does not match length 512, with accuracies of 62.9% on CASP12, 67.2% on TS115, and 64.7% on CB513. These trends suggest a trade-off: allowing moderately longer contexts reduces truncation-induced information loss, but very long sequences can introduce additional irrelevant context or noise during optimization, which may partially offset the benefits of retaining more residues.

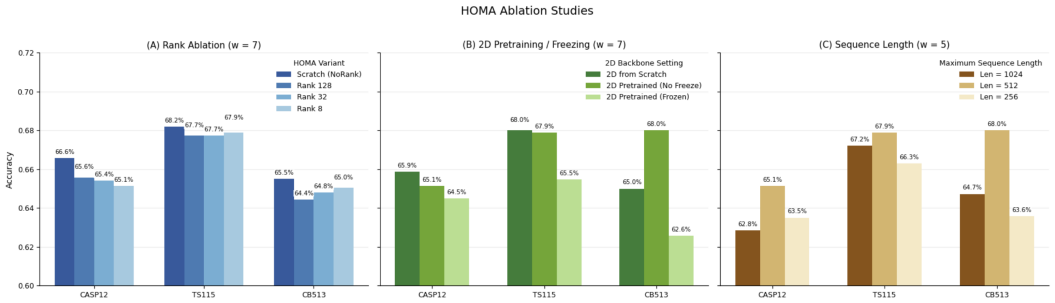

Figure 2: **HOMA ablations across datasets.** Will be rewritten. (A) Effect of low-rank setting (NoRank vs ranks 128/32/8) under $w = 7$. (B) Effect of 2D backbone initialization and freezing under $w = 7$. (C) Effect of maximum sequence length under $w = 5$. Bars report accuracy on CASP12, TS115, and CB513; y-axis is shared across panels for comparability.

5 Limitations and Broader Considerations

although performance is imporoved, there are considerations around compute and memory efficiency that can be of concern and can be further improved. optimization methods like FASTAttention can be rewritten for HOMA attention to improve the efifciency. should be applied to other fields and

domains see how applicable it is acrooss different fields. how could it be heplful for CV or NLP? Need to explain why we limited our work to only protein sequences

training with more gpu doing pretraining on larger corpus and then finetuning. Other domains seem to be promising including image analysis. write cuda kernals for it to optimize it furhter. ...

6 Conclusion

...

References

[1] Roshan Rao, Nicholas Bhattacharya, Neil Thomas, Yan Duan, Xi Chen, John Canny, Pieter Abbeel, and Yun S Song. Evaluating protein transfer learning with tape. In *Advances in Neural Information Processing Systems*, 2019.

[2] Michael Schantz Klausen, Martin Closter Jespersen, Henrik Nielsen, Kamilla Kjaergaard Jensen, Vanessa Isabell Jurtz, Casper Kaae Soenderby, Morten Otto Alexander Sommer, Ole Winther, Morten Nielsen, Bent Petersen, et al. Netsurfp-2.0: Improved prediction of protein structural features by integrated deep learning. *Proteins: Structure, Function, and Bioinformatics*, 2019.

[3] James A. Cuff and Geoffrey J. Barton. Evaluation and improvement of multiple sequence methods for protein secondary structure prediction. *Proteins: Structure, Function, and Bioinformatics*, 34(4):508–519, 1999.

[4] John Moult, Krzysztof Fidelis, Andriy Kryshtafovych, Torsten Schwede, and Anna Tramontano. Critical assessment of methods of protein structure prediction (CASP)-Round XII. *Proteins: Structure, Function, and Bioinformatics*, 86:7–15, 2018.

[5] Yuedong Yang, Jianzhao Gao, Jihua Wang, Rhys Heffernan, Jack Hanson, Kuldip Paliwal, and Yaoqi Zhou. Sixty-five years of the long march in protein secondary structure prediction: the final stretch? *Briefings in Bioinformatics*, 19(3):482–494, 2018.

[6] Karen S Sarkisyan, Dmitry A Bolotin, Margarita V Meer, Dinara R Usmanova, Alexander S Mishin, George V Sharonov, Dmitry N Ivankov, Nina G Bozhanova, Mikhail S Baranov, Onuralp Soylemez, et al. Local fitness landscape of the green fluorescent protein. *Nature*, 533(7603):397, 2016.

[7] Gabriel J Rocklin, Tamuka M Chidyausiku, Inna Goreshnik, Alex Ford, Scott Houliston, Alexander Lemak, Lauren Carter, Rashmi Ravichandran, Vikram K Mulligan, Aaron Chevalier, et al. Global analysis of protein folding using massively parallel design, synthesis, and testing. *Science*, 357(6347):168–175, 2017.

[8] IUPAC-IUB Joint Commission on Biochemical Nomenclature. Nomenclature and symbolism for amino acids and peptides (recommendations 1983). *Pure and Applied Chemistry*, 56(5):595–624, 1984.

[9] The UniProt Consortium. Uniprot: a worldwide hub of protein knowledge. *Nucleic Acids Research*, 47(D1):D506–D515, 2019.

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First-level headings should be in 12-point type.

9.1 Headings: second level

Second-level headings should be in 10-point type.

9.1.1 Headings: third level

Third-level headings should be in 10-point type.

Paragraphs There is also a `\paragraph` command available, which sets the heading in bold, flush left, and inline with the text, with the heading followed by 1 em of space.

10 Citations, figures, tables, references

These instructions apply to everyone.

10.1 Citations within the text

The `natbib` package will be loaded for you by default. Citations may be author/year or numeric, as long as you maintain internal consistency. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

The documentation for `natbib` may be found at

`http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf`

Of note is the command `\citet`, which produces citations appropriate for use in inline text. For example,

```
\citet{hasselmo} investigated\dots
```

produces

Hasselmo, et al. (1995) investigated…

If you wish to load the `natbib` package with options, you may add the following before loading the `neurips_2024` package:

```
\PassOptionsToPackage{options}{natbib}
```

If `natbib` clashes with another package you load, you can add the optional argument `nonatbib` when loading the style file:

```
\usepackage[nonatbib]{neurips_2024}
```

As submission is double blind, refer to your own published work in the third person. That is, use "In the previous work of Jones et al. [4]," not "In our previous work [4]." If you cite your other papers that are not widely available (e.g., a journal paper under review), use anonymous author names in the citation, e.g., an author of the form "A. Anonymous" and include a copy of the anonymized paper in the supplementary material.

10.2 Footnotes

Footnotes should be used sparingly. If you do require a footnote, indicate footnotes with a number[1] in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).

Note that footnotes are properly typeset *after* punctuation marks.[2]

Figure 3: Sample figure caption.

Table 3: Sample table title

	Part	
Name	Description	Size (μm)
Dendrite	Input terminal	$\sim$100
Axon	Output terminal	$\sim$10
Soma	Cell body	up to 10^6

10.3 Figures

All artwork must be neat, clean, and legible. Lines should be dark enough for purposes of reproduction. The figure number and caption always appear after the figure. Place one line space before the figure caption and one line space after the figure. The figure caption should be lower case (except for first word and proper nouns); figures are numbered consecutively.

You may use color figures. However, it is best for the figure captions and the paper body to be legible if the paper is printed in either black/white or in color.

10.4 Tables

All tables must be centered, neat, clean and legible. The table number and title always appear before the table. See Table 3.

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for first word and proper nouns); tables are numbered consecutively.

Note that publication-quality tables *do not contain vertical rules.* We strongly suggest the use of the `booktabs` package, which allows for typesetting high-quality, professional tables:

`https://www.ctan.org/pkg/booktabs`

This package was used to typeset Table 3.

10.5 Math

Note that display math in bare TeX commands will not create correct line numbers for submission. Please use LaTeX (or AMSTeX) commands for unnumbered display math. (You really shouldn't be using $$ anyway; see `https://tex.stackexchange.com/questions/503/why-is-preferable-to` and `https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath` for more information.)

[1]Sample of the first footnote.

[2]As in this example.

10.6 Final instructions

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes (except perhaps in the **References** section; see below). Please note that pages should be numbered.

11 Preparing PDF files

Please prepare submission files with paper size "US Letter," and not, for example, "A4."

Fonts were the main cause of problems in the past years. Your PDF file must only contain Type 1 or Embedded TrueType fonts. Here are a few instructions to achieve this.

- You should directly generate PDF files using `pdflatex`.
- You can check which fonts a PDF files uses. In Acrobat Reader, select the menu Files>Document Properties>Fonts and select Show All Fonts. You can also use the program `pdffonts` which comes with `xpdf` and is available out-of-the-box on most Linux machines.
- `xfig` "patterned" shapes are implemented with bitmap fonts. Use "solid" shapes instead.
- The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS Fonts:

  ```
  \usepackage{amsfonts}
  ```

  followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also use the following workaround for reals, natural and complex:

  ```
  \newcommand{\RR}{I\!\!R} %real numbers
  \newcommand{\Nat}{I\!\!N} %natural numbers
  \newcommand{\CC}{I\!\!\!\!C} %complex numbers
  ```

  Note that `amsfonts` is automatically loaded by the `amssymb` package.

If your file contains type 3 fonts or non embedded TrueType fonts, we will ask you to fix it.

11.1 Margins in LaTeX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below:

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

See Section 4.4 in the graphics bundle documentation (`http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf`)

A number of width problems arise when LaTeX cannot properly hyphenate a line. Please give LaTeX hyphenation hints using the `\-` command when necessary.

Acknowledgments and Disclosure of Funding

Use unnumbered first level headings for the acknowledgments. All acknowledgments go at the end of the paper before the list of references. Moreover, you are required to declare funding (financial activities supporting the submitted work) and competing interests (related financial activities outside the submitted work). More information about this disclosure can be found at: `https://neurips.cc/Conferences/2024/PaperInformation/FundingDisclosure`.

Do **not** include this section in the anonymized submission, only in the final paper. You can use the `ack` environment provided in the style file to automatically hide this section in the anonymized submission.

References

References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level heading for the references. Any choice of citation style is acceptable as long as you are consistent. It is permissible to reduce the font size to `small` (9 point) when listing the references. Note that the Reference section does not count towards the page limit.

[1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp. 609–616. Cambridge, MA: MIT Press.

[2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the GEneral NEural SImulation System.* New York: TELOS/Springer–Verlag.

[3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

A Appendix / supplemental material

Optionally include supplemental material (complete proofs, additional experiments and plots) in appendix. All such materials **SHOULD be included in the main submission.**